

Data Warehouse Design

Analytical Requirements

The Data Warehouse is designed to support the analysis of historical card transaction activity. It should answer the following business questions:

1. Which customer age groups generate the highest transaction value and number of transactions in different merchant categories?

Business value: This helps identify relevant customer segments for targeted offers, rewards, and marketing campaigns.

2. In which regions and merchant categories is card spending highest?

Business value: This helps identify strong markets and attractive merchant categories for promotions and potential partnerships.

3. How do transaction value and number of transactions change across months, quarters and years?

Business value: This reveals spending trends and seasonality and supports campaign planning and performance comparison over time.

4. Which card types generate the highest transaction value and usage, and how does their performance change over time?

Business value: This helps evaluate the performance and adoption of different card products and supports product development and card strategy.

Multidimensional ER Model (mER)

Fact: Transaction.

Grain: One fact row represents one individual card transaction.

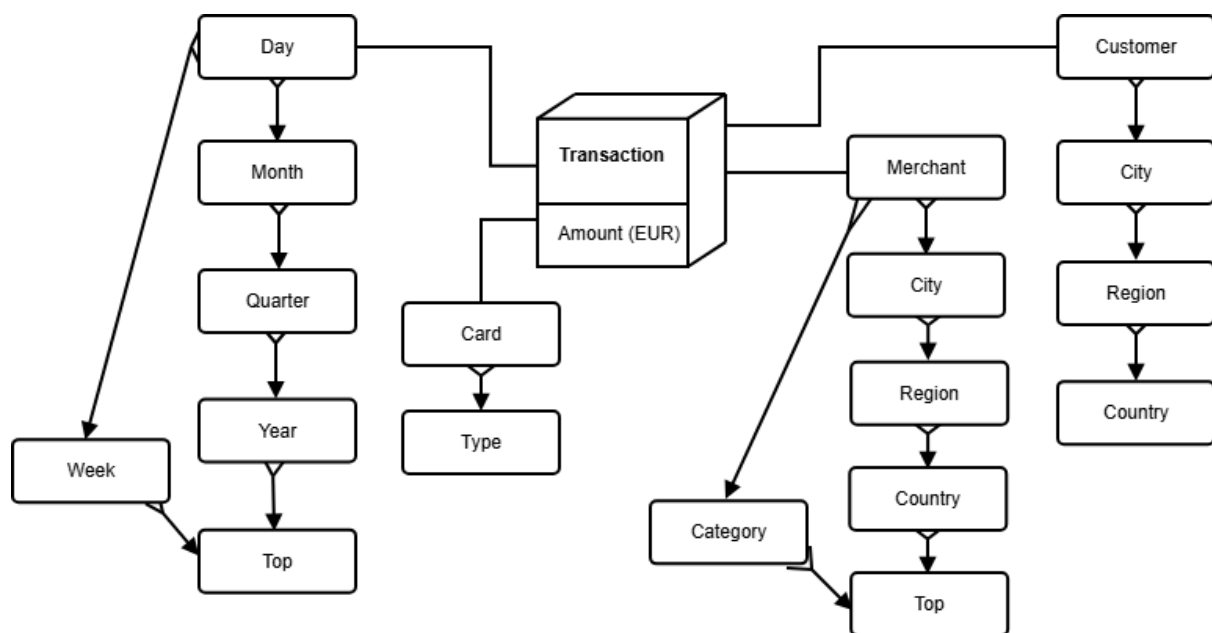
The Transaction fact is the central fact because the analytical requirements focus on transaction value and transaction volume across customers, merchants, card types and time.

Measures:

The Transaction fact contains the following measure:

- Amount (EUR): The monetary value of an individual card transaction, converted to EUR during the ETL process.

Based on the atomic transaction grain, analytical values such as total transaction value, number of transactions and average transaction value can be derived using `SUM(Amount)`, `COUNT(*)` and `AVG(Amount)`.



Logical Star Schema

The conceptual multidimensional model was transformed into a logical Star Schema. The central fact table is Transaction, with a grain of one individual card transaction. It is connected to the Time, Customer, Card, and Merchant dimensions through foreign keys. TransactionAmountEUR represents the monetary measure used for the analysis.

For the entity dimensions, source identifiers are retained alongside the Data Warehouse identifiers to preserve traceability to the source system.

